Управление процессами и заданиями

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Вывод 13

Цели и задачи работы

Цель лабораторной работы

Получить навыки работы с журналами мониторинга различных событий в системе Linux, используя службы rsyslog и systemd-journald.

Процесс выполнения лабораторной работы

Мониторинг системных событий

-

```
root@localhost:/home/manieyvazi# tail -n 20 /var/log/secure
Jan 17 16:52:13 localhost gdm-password][2482]: gkr-pam: unable to locate daemon contr
ol file
Jan 17 16:52:13 localhost gdm-password][2482]: gkr-pam: stashed password to try later
 in open session
Jan 17 16:52:13 localhost (systemd)[2492]: pam_unix(systemd-user:session): session op
ened for user manieyvazi(uid=1000) by manieyvazi(uid=0)
Jan 17 16:52:13 localhost gdm-password][2482]: pam_unix(gdm-password:session): sessio
n opened for user manieyvazi(uid=1000) by manieyvazi(uid=0)
Jan 17 16:52:13 localhost gdm-password][2482]: gkr-pam: gnome-keyring-daemon started
properly and unlocked keyring
Jan 17 16:52:24 localhost gdm-launch-environment][1348]: pam_unix(gdm-launch-environm
ent:session): session closed for user gdm
Jan 17 16:53:49 localhost (systemd)[3838]: pam_unix(systemd-user:session): session op
ened for user root(uid=0) by root(uid=0)
Jan 17 16:53:49 localhost su[3822]: pam_unix(su:session): session opened for user roo
t(uid=0) by manieyvazi(uid=1000)
Jan 17 17:15:32 localhost gdm-password][6722]: gkr-pam: unlocked login keyring
Jan 17 17:17:40 localhost su[7016]: pam_unix(su:session): session opened for user roo
t(uid=0) by manieyvazi(uid=1000)
Jan 17 17:26:01 localhost su[7016]: pam_unix(su:session): session closed for user roo
t
Jan 17 17:26:12 localhost su[3822]: pam_unix(su:session): session closed for user roo
t
Jan 17 17:26:34 localhost (systemd)[8450]: pam_unix(systemd-user:session): session op
ened for user root(uid=0) by root(uid=0)
Jan 17 17:26:34 localhost su[8433]: pam_unix(su:session): session opened for user roo
t(uid=0) by manieyvazi(uid=1000)
Jan 17 17:26:46 localhost su[8516]: pam_unix(su:session): session opened for user roo
t(uid=0) by manieyvazi(uid=1000)
Jan 17 17:26:54 localhost su[8563]: pam_unix(su:session): session opened for user roo
t(uid=0) by manieyvazi(uid=1000)
Jan 17 17:30:36 localhost su[9091]: pam_unix(su:session): session opened for user man
ieyvazi(uid=1000) by manieyvazi(uid=0)
Jan 17 17:30:42 localhost su[9128]: pam_unix(su:session): session opened for user roo
t(uid=0) by manieyvazi(uid=1000)
Jan 17 17:30:50 localhost su[9128]: pam_unix(su:session): session closed for user roo
t
Jan 17 17:30:55 localhost su[9185]: pam_unix(su:session): session opened for user roo
t(uid=0) by manieyvazi(uid=1000)
root@localhost:/home/manieyvazi#
```

Рис. 1 — Мониторинг /var/log/messages в реальном времени

Ошибки аутентификации su

-.

```
 start_thread (libc.so.6 + 0x94b68)
                                                                    #5  0x00007f04eb5066bc
 __clone3 (libc.so.6 + 0x1056bc)

                                                                    Stack trace of thread
10652:
                                                                    #0  0x00007f04eb5044bd
 syscall (libc.so.6 + 0x1034bd)
                                                                    #1  0x00000000004347a2
 n/a (n/a + 0x0)
                                                                    #2  0x00000000004506c6
 n/a (n/a + 0x0)
                                                                    #3  0x0000000000405123
 n/a (n/a + 0x0)
                                                                    #4  0x00007f04eb42b30e
 __libc_start_call_main (libc.so.6 + 0x2a30e)
                                                                    #5  0x00007f04eb42b3c9
 __libc_start_main@@GLIBC_2.34 (libc.so.6 + 0x2a3c9)
                                                                    #6  0x00000000004044aa
 n/a (n/a + 0x0)
                                                                    ELF object binary arch
itecture: AMD x86-64
Jan 17 17:37:30 localhost.localdomain systemd[1]: systemd-coredump@481-10656-0.servic
e: Deactivated successfully.
Jan 17 17:37:32 localhost.localdomain PackageKit[10208]: update-packages transaction
/544_cbcaebdc from uid 1000 finished with success after 11266ms
Jan 17 17:37:35 localhost.localdomain kernel: traps: VBoxClient[10667] trap int3 ip:4
1db4b sp:7f04dcdb4cd0 error:0 in VBoxClient[1db4b,400000+bb000]
Jan 17 17:37:35 localhost.localdomain systemd-coredump[10668]: Process 10664 (VBoxCli
ent) of user 1000 terminated abnormally with signal 5/TRAP, processing...
Jan 17 17:37:35 localhost.localdomain systemd[1]: Started systemd-coredump@482-10668-
0.service - Process Core Dump (PID 10668/UID 0).
Jan 17 17:37:35 localhost.localdomain systemd-coredump[10669]: [↗] Process 10664 (VBo
xClient) of user 1000 dumped core.

                                                                    Module libXau.so.6 fro
m rpm libXau-1.0.11-8.el10.x86_64
                                                                    Module libxcb.so.1 fro
m rpm libxcb-1.17.0-3.el10.x86_64
                                                                    Module libX11.so.6 fro
m rpm libX11-1.8.10-1.el10.x86_64
                                                                    Module libffi.so.8 fro
m rpm libffi-3.4.4-9.el10.x86_64
```

Рис. 2 — Сообщения об ошибках su в журнале

Использование команды logger

•

```
root@localhost:/home/manieyvazi# systemctl restart rsyslog.service
root@localhost:/home/manieyvazi# systemctl restart httpd
Job for httpd.service failed because the control process exited with error code.
See "systemctl status httpd.service" and "journalctl -xeu httpd.service" for details.
root@localhost:/home/manieyvazi# systemctl restart rsyslog.service
root@localhost:/home/manieyvazi# tail -f /var/log/messages-debug
Jan 17 17:47:03 localhost kernel: traps: VBoxClient[12493] trap int3 ip:41db4b sp:7f04dc
db4cd0 error:0 in VBoxClient[1db4b,400000+bb000]
Jan 17 17:47:03 localhost systemd-coredump[12494]: Process 12490 (VBoxClient) of user 10
00 terminated abnormally with signal 5/TRAP, processing...
Jan 17 17:47:03 localhost systemd[1]: Started systemd-coredump@592-12494-0.service - Pro
cess Core Dump (PID 12494/UID 0).
Jan 17 17:47:03 localhost systemd-coredump[12495]: Process 12490 (VBoxClient) of user 10
00 dumped core.#012#012Module libXau.so.6 from rpm libXau-1.0.11-8.el10.x86_64#012Module
 libxcb.so.1 from rpm libxcb-1.17.0-3.el10.x86_64#012Module libX11.so.6 from rpm libX11-
1.8.10-1.el10.x86_64#012Module libffi.so.8 from rpm libffi-3.4.4-9.el10.x86_64#012Module
 libwayland-client.so.0 from rpm wayland-1.23.0-2.el10.x86_64#012Stack trace of thread 1
2493:#012#0  0x000000000041db4b n/a (n/a + 0x0)#012#1  0x000000000041dac4 n/a (n/a + 0x0
)#012#2  0x0000000000450a7c n/a (n/a + 0x0)#012#3  0x0000000000435890 n/a (n/a + 0x0)#01
2#4  0x00007f04eb495b68 start_thread (libc.so.6 + 0x94b68)#012#5  0x00007f04eb5066bc __c
lone3 (libc.so.6 + 0x1056bc)#012#012Stack trace of thread 12490:#012#0  0x00007f04eb5044
bd syscall (libc.so.6 + 0x1034bd)#012#1  0x00000000004347a2 n/a (n/a + 0x0)#012#2  0x000
00000004506c6 n/a (n/a + 0x0)#012#3  0x0000000000405123 n/a (n/a + 0x0)#012#4  0x00007f0
4eb42b30e __libc_start_call_main (libc.so.6 + 0x2a30e)#012#5  0x00007f04eb42b3c9 __libc_
start_main@@GLIBC_2.34 (libc.so.6 + 0x2a3c9)#012#6  0x00000000004044aa n/a (n/a + 0x0)#0
12ELF object binary architecture: AMD x86-64
Jan 17 17:47:03 localhost systemd[1]: systemd-coredump@592-12494-0.service: Deactivated
successfully.
Jan 17 17:47:08 localhost kernel: traps: VBoxClient[12503] trap int3 ip:41db4b sp:7f04dc
db4cd0 error:0 in VBoxClient[1db4b,400000+bb000]
Jan 17 17:47:08 localhost systemd-coredump[12504]: Process 12500 (VBoxClient) of user 10
00 terminated abnormally with signal 5/TRAP, processing...
Jan 17 17:47:08 localhost systemd[1]: Started systemd-coredump@593-12504-0.service - Pro
cess Core Dump (PID 12504/UID 0).
Jan 17 17:47:08 localhost systemd-coredump[12505]: Process 12500 (VBoxClient) of user 10
00 dumped core.#012#012Module libXau.so.6 from rpm libXau-1.0.11-8.el10.x86_64#012Module
 libxcb.so.1 from rpm libxcb-1.17.0-3.el10.x86_64#012Module libX11.so.6 from rpm libX11-
1.8.10-1.el10.x86_64#012Module libffi.so.8 from rpm libffi-3.4.4-9.el10.x86_64#012Module
 libwayland-client.so.0 from rpm wayland-1.23.0-2.el10.x86_64#012Stack trace of thread 1
2503:#012#0  0x000000000041db4b n/a (n/a + 0x0)#012#1  0x000000000041dac4 n/a (n/a + 0x0
```

Рис. 3 — Добавление пользовательского сообщения в журнал

Журнал безопасности /var/log/secure

-.

```
manieyvazi@localhost:~$ su
Password:
root@localhost:/home/manieyvazi# mkdir -p /var/log/journal
root@localhost:/home/manieyvazi# chown root:systemd-journal /var/log/journal
root@localhost:/home/manieyvazi# chmod 2755 /var/log/journal
root@localhost:/home/manieyvazi# killall -USR1 systemd-journald
root@localhost:/home/manieyvazi# journalctl -b
Jan 17 16:51:22 localhost kernel: Linux version 6.12.0-55.41.1.el10_0.x86_64 (mockbuild@iad1-prod-build001.bld.equ.roc
Jan 17 16:51:22 localhost kernel: Command line: BOOT_IMAGE=(hd0,gpt2)/vmlinuz-6.12.0-55.41.1.el10_0.x86_64 root=/dev/m
Jan 17 16:51:22 localhost kernel: BIOS-provided physical RAM map:
Jan 17 16:51:22 localhost kernel: BIOS-e820: [mem 0x0000000000000000-0x000000000009fbff] usable
Jan 17 16:51:22 localhost kernel: BIOS-e820: [mem 0x000000000009fc00-0x000000000009ffff] reserved
Jan 17 16:51:22 localhost kernel: BIOS-e820: [mem 0x00000000000f0000-0x00000000000fffff] reserved
Jan 17 16:51:22 localhost kernel: BIOS-e820: [mem 0x0000000000100000-0x00000000dffeffff] usable
Jan 17 16:51:22 localhost kernel: BIOS-e820: [mem 0x00000000dfff0000-0x00000000dfffffff] ACPI data
Jan 17 16:51:22 localhost kernel: BIOS-e820: [mem 0x00000000fec00000-0x00000000fec00fff] reserved
Jan 17 16:51:22 localhost kernel: BIOS-e820: [mem 0x00000000fee00000-0x00000000fee00fff] reserved
Jan 17 16:51:22 localhost kernel: BIOS-e820: [mem 0x00000000fffc0000-0x00000000ffffffff] reserved
Jan 17 16:51:22 localhost kernel: BIOS-e820: [mem 0x0000000100000000-0x00000002203fffff] usable
Jan 17 16:51:22 localhost kernel: NX (Execute Disable) protection: active
Jan 17 16:51:22 localhost kernel: APIC: Static calls initialized
Jan 17 16:51:22 localhost kernel: SMBIOS 2.5 present.
Jan 17 16:51:22 localhost kernel: DMI: innotek GmbH VirtualBox/VirtualBox, BIOS VirtualBox 12/01/2006
Jan 17 16:51:22 localhost kernel: DMI: Memory slots populated: 0/0
Jan 17 16:51:22 localhost kernel: Hypervisor detected: KVM
Jan 17 16:51:22 localhost kernel: kvm-clock: Using msrs 4b564d01 and 4b564d00
Jan 17 16:51:22 localhost kernel: kvm-clock: using sched offset of 6081827587 cycles
Jan 17 16:51:22 localhost kernel: clocksource: kvm-clock: mask: 0xffffffffffffffff max_cycles: 0x1cd42e4dffb, max_idle
Jan 17 16:51:22 localhost kernel: tsc: Detected 2687.998 MHz processor
Jan 17 16:51:22 localhost kernel: e820: update [mem 0x00000000-0x00000fff] usable ==> reserved
Jan 17 16:51:22 localhost kernel: e820: remove [mem 0x000a0000-0x000fffff] usable
Jan 17 16:51:22 localhost kernel: last_pfn = 0x220400 max_arch_pfn = 0x400000000
Jan 17 16:51:22 localhost kernel: MTRR map: 3 entries (3 fixed + 0 variable; max 19), built from 8 variable MTRRs
Jan 17 16:51:22 localhost kernel: x86/PAT: Configuration [0-7]: WB  WC  UC- UC  WB  WP  UC- WT
Jan 17 16:51:22 localhost kernel: CPU MTRRs all blank - virtualized system.
Jan 17 16:51:22 localhost kernel: last_pfn = 0xe0000 max_arch_pfn = 0x400000000
Jan 17 16:51:22 localhost kernel: found SMP MP-table at [mem 0x0009fbf0-0x0009fbff]
Jan 17 16:51:22 localhost kernel: RAMDISK: [mem 0x34340000-0x36197fff]
Jan 17 16:51:22 localhost kernel: ACPI: Early table checksum verification disabled
Jan 17 16:51:22 localhost kernel: ACPI: RSDP 0x000000000000E0000 000024 (v02 VBOX  )
Jan 17 16:51:22 localhost kernel: ACPI: XSDT 0x00000000DFFF0030 00003C (v01 VBOX   VBOXXSDT 00000001 ASL  00000061)
Jan 17 16:51:22 localhost kernel: ACPI: FACP 0x00000000DFFF00F0 0000F4 (v04 VBOX   VBOXFACP 00000001 ASL  00000061)
Jan 17 16:51:22 localhost kernel: ACPI: DSDT 0x00000000DFFF0340 002353 (v02 VBOX   VBOXBIOS 00000002 VBOX 00029BF3)
Jan 17 16:51:22 localhost kernel: ACPI: FACS 0x00000000DFFF0200 000040
Jan 17 16:51:22 localhost kernel: ACPI: FACS 0x00000000DFFF0200 000040
Jan 17 16:51:22 localhost kernel: ACPI: APIC 0x00000000DFFF0240 00007C (v02 VBOX   VBOXAPIC 00000001 ASL  00000061)
```

Рис. 4 — Просмотр последних записей журнала secure

Изменение правил rsyslog.conf

-.

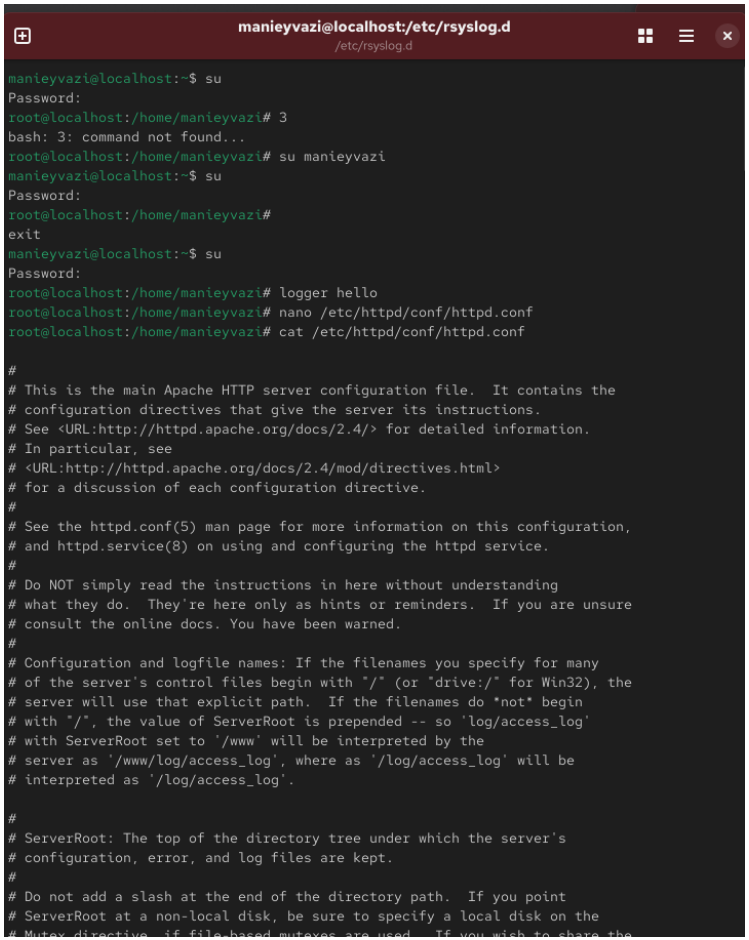

Рис. 5 — Запуск службы httpd и её журнал

Настройка веб-сервера Apache

-.

```
manieyvazi@localhost:~$ su
Password:
root@localhost:/home/manieyvazi# mkdir -p /var/log/journal
root@localhost:/home/manieyvazi# chown root:systemd-journal /var/log/journal
root@localhost:/home/manieyvazi# chmod 2755 /var/log/journal
root@localhost:/home/manieyvazi# killall -USR1 systemd-journald
root@localhost:/home/manieyvazi# journalctl -b
Jan 17 16:51:22 localhost kernel: Linux version 6.12.0-55.41.1.el10_0.x86_64 (mockbuild@iad1-prod-build001.bld.equ.roc
Jan 17 16:51:22 localhost kernel: Command line: BOOT_IMAGE=(hd0,gpt2)/vmlinuz-6.12.0-55.41.1.el10_0.x86_64 root=/dev/m
Jan 17 16:51:22 localhost kernel: BIOS-provided physical RAM map:
Jan 17 16:51:22 localhost kernel: BIOS-e820: [mem 0x0000000000000000-0x000000000009fbff] usable
Jan 17 16:51:22 localhost kernel: BIOS-e820: [mem 0x000000000009fc00-0x000000000009ffff] reserved
Jan 17 16:51:22 localhost kernel: BIOS-e820: [mem 0x00000000000f0000-0x00000000000fffff] reserved
Jan 17 16:51:22 localhost kernel: BIOS-e820: [mem 0x0000000000100000-0x00000000dffeffff] usable
Jan 17 16:51:22 localhost kernel: BIOS-e820: [mem 0x00000000dfff0000-0x00000000dfffffff] ACPI data
Jan 17 16:51:22 localhost kernel: BIOS-e820: [mem 0x00000000fec00000-0x00000000fec00fff] reserved
Jan 17 16:51:22 localhost kernel: BIOS-e820: [mem 0x00000000fee00000-0x00000000fee00fff] reserved
Jan 17 16:51:22 localhost kernel: BIOS-e820: [mem 0x00000000fffc0000-0x00000000ffffffff] reserved
Jan 17 16:51:22 localhost kernel: BIOS-e820: [mem 0x0000000100000000-0x00000002203fffff] usable
Jan 17 16:51:22 localhost kernel: NX (Execute Disable) protection: active
Jan 17 16:51:22 localhost kernel: APIC: Static calls initialized
Jan 17 16:51:22 localhost kernel: SMBIOS 2.5 present.
Jan 17 16:51:22 localhost kernel: DMI: innotek GmbH VirtualBox/VirtualBox, BIOS VirtualBox 12/01/2006
Jan 17 16:51:22 localhost kernel: DMI: Memory slots populated: 0/0
Jan 17 16:51:22 localhost kernel: Hypervisor detected: KVM
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Jan 17 16:51:22 localhost kernel: clocksource: kvm-clock: mask: 0xffffffffffffffff max_cycles: 0x1cd42e4dffb, max_idle
Jan 17 16:51:22 localhost kernel: tsc: Detected 2687.998 MHz processor
Jan 17 16:51:22 localhost kernel: e820: update [mem 0x00000000-0x00000fff] usable ==> reserved
Jan 17 16:51:22 localhost kernel: e820: remove [mem 0x000a0000-0x000fffff] usable
Jan 17 16:51:22 localhost kernel: last_pfn = 0x220400 max_arch_pfn = 0x400000000
Jan 17 16:51:22 localhost kernel: MTRR map: 3 entries (3 fixed + 0 variable; max 19), built from 8 variable MTRRs
Jan 17 16:51:22 localhost kernel: x86/PAT: Configuration [0-7]: WB  WC  UC- UC  WB  WP  UC- WT
Jan 17 16:51:22 localhost kernel: CPU MTRRs all blank - virtualized system.
Jan 17 16:51:22 localhost kernel: last_pfn = 0xe0000 max_arch_pfn = 0x400000000
Jan 17 16:51:22 localhost kernel: found SMP MP-table at [mem 0x0009fbf0-0x0009fbff]
Jan 17 16:51:22 localhost kernel: RAMDISK: [mem 0x34340000-0x36197fff]
Jan 17 16:51:22 localhost kernel: ACPI: Early table checksum verification disabled
Jan 17 16:51:22 localhost kernel: ACPI: RSDP 0x000000000000E0000 000024 (v02 VBOX  )
Jan 17 16:51:22 localhost kernel: ACPI: XSDT 0x00000000DFFF0030 00003C (v01 VBOX   VBOXXSDT 00000001 ASL  00000061)
Jan 17 16:51:22 localhost kernel: ACPI: FACP 0x00000000DFFF00F0 0000F4 (v04 VBOX   VBOXFACP 00000001 ASL  00000061)
Jan 17 16:51:22 localhost kernel: ACPI: DSDT 0x00000000DFFF0340 002353 (v02 VBOX   VBOXBIOS 00000002 VBOX 00029BF3)
Jan 17 16:51:22 localhost kernel: ACPI: FACS 0x00000000DFFF0200 000040
Jan 17 16:51:22 localhost kernel: ACPI: FACS 0x00000000DFFF0200 000040
Jan 17 16:51:22 localhost kernel: ACPI: APIC 0x00000000DFFF0240 00007C (v02 VBOX   VBOXAPIC 00000001 ASL  00000061)
```

Рис. 6 — Изменение конфигурации httpd.conf

Настройка ErrorLog syslog:local1

-.

```
#
# Load config files in the "/etc/httpd/conf.d" directory, if any.
IncludeOptional conf.d/*.conf
root@localhost:/home/manieyvazi# cd /etc/rsyslog.d
root@localhost:/etc/rsyslog.d# touch httpd.conf
root@localhost:/etc/rsyslog.d# nano httpd.conf
root@localhost:/etc/rsyslog.d# cd /etc/rsyslog.d
root@localhost:/etc/rsyslog.d# touch debug.conf
root@localhost:/etc/rsyslog.d# echo "*.debug /var/log/messages-debug" >

bash: syntax error near unexpected token `newline'
root@localhost:/etc/rsyslog.d# echo "*.debug /var/log/messages-debug" >
bash: syntax error near unexpected token `newline'
root@localhost:/etc/rsyslog.d# echo "*.debug /var/log/messages-debug" >
bash: syntax error near unexpected token `newline'
root@localhost:/etc/rsyslog.d# echo "*.debug /var/log/messages-debug" > /etc/rsyslog.d/de
bug.conf
root@localhost:/etc/rsyslog.d# logger -p daemon.debug "Daemon Debug Message"
root@localhost:/etc/rsyslog.d# 3
```

Рис. 7 — Файл /etc/rsyslog.d/httpd.conf

Создание файла правил httpd.conf

-.

```
manieyvazi@localhost:~$ su
Password:
root@localhost:/home/manieyvazi# 1
bash: 1: command not found...
root@localhost:/home/manieyvazi# dnf -y install httpd
Extra Packages for Enterprise Linux 10 - x86_64          54 kB/s |  39 kB     00:00
Extra Packages for Enterprise Linux 10 - x86_64          48 kB/s | 5.6 MB     01:57
Rocky Linux 10 - BaseOS                                 5.7 kB/s | 4.3 kB     00:00
Rocky Linux 10 - BaseOS                                 4.3 MB/s | 7.6 MB     00:01
Rocky Linux 10 - AppStream                               14 kB/s | 4.3 kB     00:00
Rocky Linux 10 - AppStream                              2.7 MB/s | 2.1 MB     00:00
Rocky Linux 10 - Extras                                  11 kB/s | 3.1 kB     00:00
Rocky Linux 10 - Extras                                 6.8 kB/s | 5.9 kB     00:00
Dependencies resolved.
================================================================================
 Package                 Architecture  Version                 Repository       Size
================================================================================
Installing:
 httpd                   x86_64        2.4.63-4.el10_1.3       appstream        52 k
Installing dependencies:
 apr                     x86_64        1.7.5-2.el10            appstream       128 k
 apr-util                x86_64        1.6.3-21.el10           appstream        98 k
 apr-util-lmdb           x86_64        1.6.3-21.el10           appstream        14 k
 httpd-core              x86_64        2.4.63-4.el10_1.3       appstream       1.5 M
 httpd-filesystem        noarch        2.4.63-4.el10_1.3       appstream        14 k
 httpd-tools             x86_64        2.4.63-4.el10_1.3       appstream        81 k
 rocky-logos-httpd       noarch        100.4-7.el10            appstream        24 k
Installing weak dependencies:
 apr-util-openssl        x86_64        1.6.3-21.el10           appstream        16 k
 mod_http2               x86_64        2.0.29-3.el10           appstream       164 k
 mod_lua                 x86_64        2.4.63-4.el10_1.3       appstream        59 k

Transaction Summary
================================================================================
Install  11 Packages

Total download size: 2.1 M
Installed size: 6.1 M
Downloading Packages:
(1/11): apr-util-lmdb-1.6.3-21.el10.x86_64.rpm          162 kB/s |  14 kB     00:00
(2/11): apr-util-1.6.3-21.el10.x86_64.rpm               770 kB/s |  98 kB     00:00
(3/11): apr-util-openssl-1.6.3-21.el10.x86_64.rpm       359 kB/s |  16 kB     00:00
(4/11): apr-1.7.5-2.el10.x86_64.rpm                     902 kB/s | 128 kB     00:00
(5/11): httpd-2.4.63-4.el10_1.3.x86_64.rpm              739 kB/s |  52 kB     00:00
```

Рис. 8 — Файл /etc/rsyslog.d/debug.conf

Создание файла debug.conf

Рис. 9 — Сообщение от logger в messages-debug

Проверка записи отладочных сообщений

-

Рис. 10 — Просмотр системного журнала

Выводы по проделанной работе

Вывод

В ходе лабораторной работы были изучены принципы системного журналирования в Linux. Были рассмотрены возможности служб rsyslog и systemd-journald, настройка пользовательских правил, регистрация ошибок и отладочной информации. Освоены команды фильтрации и анализа логов с помощью journalctl, а также способы организации постоянного хранения журналов для долгосрочного мониторинга системы.